Exp.No: 7
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Sliding Window

Aim:

To write sliding window protocol code using python.

Code

sender.py

```
import time
import os
ws = int(input("Window size"))
msg = input("Enter message:")
frames = [(i+1, ch) for i, ch in enumerate(msg)]
i=0
while i< len(frame):
    windows = frames[i : i+ws]
    with open("sender_Buffer", "w") as f:
        for nodata in window:
            f.write(f"{no}, {data}\n")
    print("\n Sending :" window)
    time.sleep(2)
    while not os.path.exists("Receiver Buffer"):
        time.sleep(1)
    with open("Receiver-Buffer") as f:
        ack, no = f.read().strip().split();
        no = int(no)
    print("Received", ack, no)
    if ack == "Ack" and no == window[-1][0]:
        i += ws
    else ack == "NACK":
        i = no-1
```

```
        open("Receiver_Buffer", "w").close()
print("\n Transmission Completed")
```

receiver.py

```
import time,os
expected = 1
while True:
    if not os.path.exists("sender_Buffer"):
        time.sleep(1)
        continue
    with open("sender_Buffer") as f:
        frames = [(int(x.split(",")[0]), x.split(",")[1].
                   strip() for x in f if x.strip()]
    if not frames:
        time.sleep(1)
        continue
    print("\n Received", frames)
    for no, data in frames:
        if no != expected:
            with open("Receiver_Buffer", "w") as f:
                f.write(f"NAC, {expected}")
            break
        expected += 1
    else:
        with open("Receiver_Buffer", "w") as f:
            f.write(f"Ack {frame[-1][0]}")
    time.sleep(1)
```

O/p

Sender terminal

Window Size : 3

Enter message : BRICKS

Sending [(1,'B'),(2,'R'),(3,'I')]

Received : ACK 3

Sending [(4,'K')(5,'S')]

Received ACK 4

Transmission completed

Receiver Terminal

Received : [(1,'B'),(2,'R'),(3,'C')]

Received : [(4,'K')(5,'S')]

Result:

Thus the code is implemented successfully for Sliding Window.